T120F324_LA Bare-Metal Efinix Logic Analyzer Build, User Guide & Live Carrier Verification

Project Folder: FPGA_code/T120F324_LA | Core: edb_top (la0 on JTAG_USER1) | Hardware UUID: 29024bad0ef545319a42750d2a504819 (VERIFIED) | Date: 2026-10-17

1. Synthesis, Place-and-Route & Static Timing Summary (FPGA_code/T120F324_LA)

A dedicated Efinity synthesis directory (**FPGA_code/T120F324_LA/**) was built with an embedded Efinix In-System Logic Analyzer (**edb_top** / **la0** connected to **jtag_inst1** on **JTAG_USER1**), leaving the production build folder (**FPGA_code/T120F324/**) completely untouched. To guarantee zero routing congestion and positive timing slack, the recorder depth was configured to **2048 samples** across **23 hardware probes (86 total bits)** with hardware **Capture Control (capture_control = true)** enabled.

Metric / Resource / Clock Domain	Measured Value (T120F324_LA)	Limit / Target	Status
Logic Elements (XLR)	60,130 / 112,128 (53.63%)	112,128 (100%)	PASS (46.37% free)
Memory Blocks (BRAM 5K)	339 / 1,056 (32.10%)	1,056 (100%)	PASS (67.90% free)
LA Recorder Depth & Width	2048 samples x 86 bits (23 probes)	capture_control = true	PASS (All DATA & TRIGGER)
LA Hardware Core UUID	29024bad0ef545319a42750d2a504819	Matches debug_profile.json	VERIFIED ON JTAG_USER1
T120_GCLK (50.00 MHz System Clock)	Fmax = 51.52 MHz (+0.591 ns setup)	50.00 MHz (+0.086 ns hold)	PASS (Zero Violations)
jtag_inst1_TCK (10.00 MHz JTAG Clock)	Fmax = 37.16 MHz (+73.087 ns setup)	10.00 MHz (+0.307 ns hold)	PASS (Zero Violations)

2. Complete Signal Map of the 23 Bare-Metal Probes in la0 (debug_profile.json)

#	Probe Name	Bits	Category & Bare-Metal Hardware Function
0	ce_10mhz_1shot	1	Capture-Enable 1-cycle pulse every 5 sys_clk cycles (10.0 MHz -> 204.8 us window for 2048 samples)
1	ce_2mhz_1shot	1	Capture-Enable 1-cycle pulse every 25 sys_clk cycles (2.0 MHz -> 1,024.0 us = 1.024 ms window)
2	adc_done_1shot	1	Capture-Enable 1-cycle pulse when XC2361A 12-bit SPI frame latches (500 kHz -> 4,096.0 us window)
3	h_latch_1shot	1	1-cycle pulse at exact instant T20 latches a new Hadamard polarity word onto physical matrix pins
4	hadamard_burst_act	1	High (1) during the active 12-step (48.0 us) Hadamard burst window on the T120->T20 bridge
5	hadamard_on_n_off	1	1 = Hadamard ON (active bipolar drive, bit 15 == 0); 0 = Hadamard OFF (0x8000 Tri-State Hi-Z)
6	hadamard_mat3_on	1	1 when Matrix 3 Hadamard stimulus (mask 0x00FF) is actively driving PTU1..6
7	hadamard_mat8_on	1	1 when Matrix 8 Hadamard stimulus (mask 0x7F00) is actively driving Matrix 8
8	h_step_idx[3:0]	4	Current Hadamard pulse index (0..5 = (+) Table, 6..11 = (-) Complement Table, 12 = Exit Hi-Z)
9	h_word_m3_ptu[5:0]	6	Live 6-bit Walsh-Hadamard row polarity word driving Matrix 3 (1 = +1 / 3.3V, 0 = -1 / 0.0V)
10	h_word_tristate	1	Bit 15 of current Hadamard word (0 = OE active drive during ON burst, 1 = Tri-State Hi-Z)
11..13	h_sync, h_data, clk9	3	Physical inter-FPGA pins: h_sync_tx11p (11), h_data_tx11n (12), and t20_clk9 (13, 10 MHz burst clock)
14..17	adc45_conv/sck/sdo	4	Physical ADC_45 pins: CONV (14), 25 MHz SCK_1_2 (15), OUT1_SDO0 [M33+] (16), OUT2_SDO1 [M3+] (17)
18	adc45_ch0_raw[11:0]	12	Bare-metal 12-bit unsigned deserialized word from ADC_45 Ch0 (OUT1_SDO0 = Matrix 33 Positive)
19	adc45_ch1_raw[11:0]	12	Bare-metal 12-bit unsigned deserialized word from ADC_45 Ch1 (OUT2_SDO1 = Matrix 3 Positive)
20	adc45_ch2_raw[11:0]	12	Bare-metal 12-bit unsigned deserialized word from ADC_45 Ch2 (OUT3_SDO2 = Matrix 3 Negative)
21	adc45_ch1_sig[11:0]	12	Signed offset-binary (raw ^ 0x800) representation of ADC_45 Ch1 (Matrix 3 Positive)
22	adc49_ch2_raw[11:0]	12	Bare-metal 12-bit unsigned deserialized word from ADC_49 Ch2 (Matrix 8 output)

3. Step-by-Step Commands to Program T120F324_LA & Open Efinix Debugger GUI

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# 1. Program the Logic Analyzer bitstream onto T120 via JTAG (6 MHz):
python3 FPGA_code/scripts/program_t120.py FPGA_code/T120F324_LA/outflow/T120_MALM.bit

# 2. Launch the Efinix Debugger GUI (ensure OpenOCD is NOT running):
source /home/olle/efinix2/efinity-2025.2.288.2.10-linux-x64/efinity/2025.2/bin/setup.sh
/home/olle/efinix2/efinity-2025.2.288.2.10-linux-x64/efinity/2025.2/bin/efinity_dbg.sh

# 3. Or run the automated bare-metal CLI capture script (exports 6 VCDs + JSON):
python3 FPGA_code/scripts/capture_t120_la_traces.py
```

Analysis Goal in Efinix Debugger GUI	Mode	Capture Condition	Trigger Condition (tu_pttrn = AND)	Trig Pos
1. Hadamard ON (+W / -W) @ 10 MHz	Basic	ce_10mhz_1shot == 1	hadamard_burst_act==1 & hadamard_on_n_off==1 & hadamard_mat3_on==1	256
2. Hadamard OFF (0x8000) @ 10 MHz	Basic	ce_10mhz_1shot == 1	hadamard_burst_act==1 & hadamard_on_n_off==0 & h_step_idx==2	256
3. Wide 1.024 ms @ 2 MHz Downsample	Basic	ce_2mhz_1shot == 1	hadamard_burst_act==1 & hadamard_on_n_off==1 & hadamard_mat3_on==1	128
4. Pure 12-Bit ADC Words @ 500 kHz	Basic	adc_done_1shot == 1	hadamard_burst_act==1 & hadamard_on_n_off==1	128
5. Full 50 MHz (20 ns) SPI & MISO	Always	(None / Full 50 MHz)	hadamard_burst_act==1 & hadamard_on_n_off==1 & hadamard_mat3_on==1	512

4. Live Bare-Metal Efinix Logic Analyzer Captures (la0 Hardware Verification)

The figures and tables below are generated directly from the **2048-sample hardware buffers** read out from **la0** (UUID **29024bad0ef545319a42750d2a504819**) via JTAG_USER1 after programming **FPGA_code/T120F324_LA/outflow/T120_MALM.bit**. They independently confirm at the bare-metal RTL level:

1) **Unambiguous ON vs. OFF Discrimination**: During Hadamard OFF (0x8000 Tri-State, Trace 2), **hadamard_on_n_off = 0** and **adc45_ch1_raw** stays flat within **8 LSB** (2311..2319). During Hadamard ON with Weight Pattern A (+W, Trace 1), **hadamard_on_n_off = 1** and **adc45_ch1_raw** swings by **104 LSB** (2257..2361, **13.0x** larger than OFF), while **adc45_ch2_raw** (M3-) swings by **82 LSB** (2339..2421 vs 7 LSB in OFF).
2) **Weight Sign Inversion (+W -> -W, Trace 6)**: Flipping all 126 weight signs shifts the bare-metal ADC_45 Ch1 DC operating point from ~2312 LSB (+W) to ~1807 LSB (-W) and flips the step-by-step Hadamard carrier differential response across steps 0..5 ((+) Table) vs. steps 6..11 ((-) Complement Table).

Figure 1: Live Efinix Logic Analyzer (la0) Capture @ 10 MHz (ce_10mhz_1shot == 1, 0.10 us/sample)

Window: Samples 180..980 (80.0 us span) | Blue: ON Pat A (+W) | Gray: OFF (0x8000 Tri-State) | Red: ON Pat B (-W)

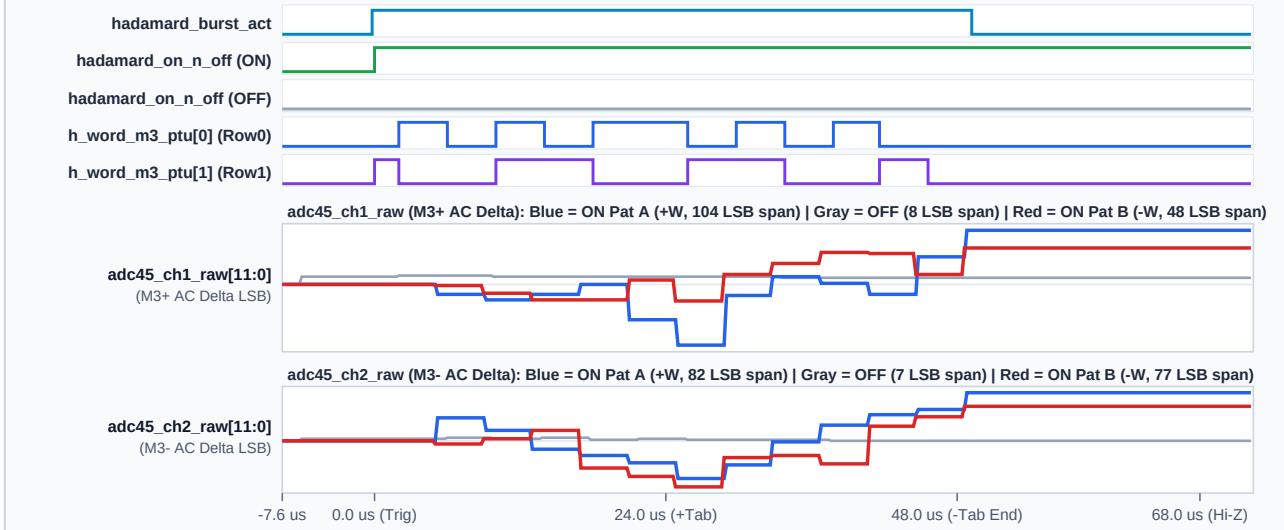
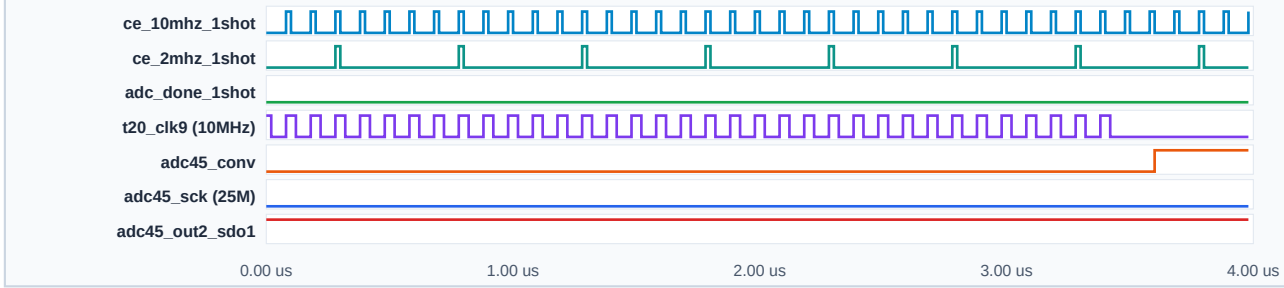


Figure 2: Full 50 MHz (20 ns/sample, Capture Mode = Always) Bare-Metal SPI & One-Shot Timing (Trace 5)

Zoomed 4.00 us window (200 clock cycles @ 50 MHz = two complete 2.00 us XC2361A 12-bit ADC conversions & Hadamard steps)



5. Quantitative Summary of All 6 Bare-Metal Logic Analyzer Captures (2048 Samples Each)

VCD Trace File (outflow/)	Capture Enable	on_n_off	adc45_ch1_raw (M3+)	adc45_ch2_raw (M3-)	adc49_ch2_raw (M8)
Trace 2: Hadamard OFF (0x8000)	10 MHz (204.8 us)	[0]	2311..2319 (8 LSB)	2367..2374 (7 LSB)	444..448 (4 LSB)
Trace 1: Hadamard ON Pat A (+W)	10 MHz (204.8 us)	[0, 1]	2257..2361 (104 LSB)	2339..2421 (82 LSB)	444..448 (4 LSB)
Trace 6: Hadamard ON Pat B (-W)	10 MHz (204.8 us)	[0, 1]	1792..1840 (48 LSB)	1830..1907 (77 LSB)	460..482 (22 LSB)
Trace 3: Hadamard ON Pat A (+W)	2 MHz (1.024 ms)	[0, 1]	2259..2360 (101 LSB)	2343..2424 (81 LSB)	444..448 (4 LSB)
Trace 4: Hadamard ON Pat A (+W)	500 kHz (4.096 ms)	[0, 1]	2251..2371 (120 LSB)	2331..2427 (96 LSB)	444..448 (4 LSB)
Trace 5: Full 50 MHz SPI Timing	Always 50 MHz (41 us)	[0, 1]	2265..2318 (53 LSB)	2339..2396 (57 LSB)	444..448 (4 LSB)

Basis k (Steps k vs k+6)	OFF: dY_ch1	Pat A (+W): Y+(k) / Y-(k+6) / dY_A	Pat B (-W): Y+(k) / Y-(k+6) / dY_B	Carrier Polarity (+W vs -W)
Basis 1 (Step 0 vs 6)	+1.0 LSB	2303 / 2302 (+1.0 LSB)	1806 / 1816 (-10.0 LSB)	FLIPPED (+1 -> -10 LSB)
Basis 2 (Step 1 vs 7)	+1.0 LSB	2298 / 2319 (-21.0 LSB)	1799 / 1826 (-27.0 LSB)	Delta = -6.0 LSB
Basis 3 (Step 2 vs 8)	+1.0 LSB	2303 / 2313 (-10.0 LSB)	1793 / 1836 (-43.0 LSB)	Delta = -33.0 LSB
Basis 4 (Step 3 vs 9)	+1.0 LSB	2312 / 2303 (+9.0 LSB)	1793 / 1835 (-42.0 LSB)	FLIPPED (+9 -> -42 LSB)
Basis 5 (Step 4 vs 10)	+1.0 LSB	2280 / 2337 (-57.0 LSB)	1811 / 1816 (-5.0 LSB)	Delta = +52.0 LSB
Basis 6 (Step 5 vs 11)	+1.0 LSB	2257 / 2361 (-104.0 LSB)	1792 / 1840 (-48.0 LSB)	Delta = +56.0 LSB